Table 3.7 Relationship between word frequency and relevance

Word frequency	Relevancy
Higher term frequency	Higher relevancy
Higher document frequency	Lower relevancy

Stop words

Words such as *the, a, it, an, but, if, for,* and *and*, are called *stop words* and can be removed by using a stop filter plugin. The default `standard` analyzer doesn't have the `stopwords` parameter enabled (the `stopwords` filter is set to `_none_` by default), so these words are analyzed. However, if our requirement is to ignore these words, we can enable the stop words filter by adding the parameter `stopwords` set to `_english_`, as shown here:

```
PUT index_with_stopwords
{
  "settings": {
    "analysis": {
      "analyzer": {
        "standard_with_stopwords_enabled": {
          "type": "standard",
          "stopwords": "_english_"
        }
      }
    }
  }
}
```

We learn about customizing analyzers in chapter 7.

Until version 5.0, Elasticsearch used the TF-IDF similarity function to calculate scores and rank results. The TF-IDF function was deprecated in favor of the BM25 function. The TF-IDF algorithm didn't consider the field's length, which skewed the relevancy scores. For example, which of these documents do you think is more relevant to the search criteria?

- A field with 100 words, including 5 occurrences of a search word
- A field with 10 words, including 3 occurrences of the search word

Logically, it may be obvious that the second document is the most relevant as it has more search words in a shorter length. Elasticsearch improved its similarity algorithms by enhancing TF-IDF with an additional parameter: field length.

The *field-length norm* provides a score based on the length of the field: the search word occurring multiple times in a short field is more relevant. For example, the word *Java* appearing once over a long synopsis may not indicate a useful result. On the